Database Migration Protocol & Zero-Downtime Rules

Database Migration Architecture Guide: Enforcing Zero-Downtime Schema Evolutions across PostgreSQL and DynamoDB.

Principles of Zero-Downtime Migrations: All database migrations must follow the Expand and Contract pattern across multiple software releases. Direct dropping of columns or renaming tables in a single step is strictly prohibited.

Phase 1 (Expand): Add the new nullable column or secondary table. Deploy backend code that writes to both old and new schemas while reading from the old schema.

Phase 2 (Backfill): Execute an asynchronous batch backfill script during off-peak hours using our internal db-worker job queue.

Phase 3 (Switch Read): Deploy software release to read from the new schema.

Phase 4 (Contract): Remove legacy code pathways and safely deprecate the unused old column in a subsequent release cycle.